

Ans 1-1 Boys sit together and girls sit together

[BBB] [GGG]

2 Blocks can be arranged in : $2!$

within Blocks

- i Boys can be arranged : $3!$
- ii Girls can be arranged : $3!$

$$\text{Final Ans : } 2! \times 3! \times 3! = \boxed{72}$$

Ans 1-2 only Boys sit together :

[BBB], G_1, G_2, G_3

4 object can be arranged : $4!$

Arrange boys : $3!$

$$\text{Final Ans : } 4! \times 3! = \boxed{144}$$

Ans 1.3 No 2 people of Same Sex together

Possibility 1 $\rightarrow$ B G B G B G

Possibility 2 $\rightarrow$ G B G B G B

For each position :

i boys arrangement : $3!$

ii girls arrangement : $3!$

$$\text{Final Ans} = 2 \times 3! \times 3! = \boxed{72}$$

Ans 2.1 Total number of arrangement = $N!$

To count arrangement where William & Gracie are together

$$= (N-1)!$$

within block, they can appear in 2 ways

$$\text{So, Total Favourable arrangement} = 2(N-1)!$$

$$P(\text{William and Gracie adjacent}) = \frac{2(N-1)!}{N!}$$

$$= \boxed{\frac{2}{N}}$$

Ans 2.2 Total arrangement in circle = $(N-1)!$

$$P(\text{William and Gracie Sit adjacent}) = \frac{2(N-2)!}{(N-1)!}$$

$$= \frac{2}{N-1}$$

$$\Rightarrow P(\text{Not adjacent}) = 1 - \frac{2}{N-1}$$

$$= \frac{N-1-2}{N-1}$$

$$= \frac{N-3}{N-1}$$

Ans 3.1 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$\Rightarrow \frac{5}{16} = \frac{1}{8} + \frac{1}{4} - P(A \cap B)$$

$$\frac{5}{16} = \frac{3}{8} - P(A \cap B)$$

$$P(A \cap B) = \frac{1}{16}$$

Ans 3.2 The largest possible value of $P(A \cup B)$ occurs when A and B are disjoint
So $P(A \cap B) = 0$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) \\ &= \frac{1}{8} + \frac{1}{4} \\ &= \frac{3}{8} \end{aligned}$$

This is less than $\frac{1}{2}$

So it is impossible for $P(A \cup B) \geq \frac{1}{2}$

Ans 3.3 The smallest possible union occurs when smaller event is subset of larger one

So

$$A \subseteq B \quad \text{then} \quad A \cap B = A$$

So,

$$P(A \cup B) = P(B) = \frac{1}{4}$$

=

Yes, but only when $P(A \cup B) = \frac{1}{4}$

Ans 4

$$\Omega = \{0, 1, 2, \dots, n\}$$
$$P(\{K\}) = cK$$

Axiom 2 (normalization)

$$\sum_{K=0}^n cK = c \sum_{K=0}^n K$$

$$= \frac{c(n)(n+1)}{2} = 1$$

$$c = \frac{2}{n(n+1)}$$

Axiom 1 check

we need $0 \leq c_K \leq 1$ for
all $K \in \Omega$

$$P(\{K\}) = c_K = \frac{2^K}{h(h+1)}$$

because $K \geq 0$; $P(\{K\}) \geq 0$

The largest probability occurs
at $K = n$

$$P(\{n\}) = \frac{2^n}{h(h+1)} = \frac{2}{h+1}$$

for all $h \geq 1$

So both are satisfied by

$$c = \frac{2}{h(h+1)}$$

Ans 5 Red ball = n ; Blue ball = m
Total ball drawn = K

1 Among the first $K-1$ ball,
there must be exactly $r-1$ red
balls

2 The K^{th} must be red

if exactly $r-1$ of first $K-1$
balls are red then remaining

$$(K-1) - (r-1) = K-r \text{ balls}$$

among $K-1$ must
be blue

$$P(\text{first } K-1 \text{ draws}) = \frac{{}^n C_{r-1} \times {}^m C_{K-r}}{{}^{n+m} C_{K-1}}$$

After this, the urn contains

1 $n - (r-1) = n - r + 1$ red balls

2 $n + m - (K-1) = n + m - K + 1$
total balls

∴ Conditional Probability
that k^{th} ball is red

$$= \frac{n-r+1}{n+m-k+1}$$

Multiplying both

$$P(K=r) = \frac{{}^n C_{r-1} \times {}^m C_{k-r} \times \frac{n-r+1}{n+m-k+1}}{{}^{n+m} C_{k-1}}$$

$$P(K=r) = \frac{{}^{r-1} C_{r-1} \times {}^{n+m-k} C_{k-r}}{{}^{n+m} C_n}$$

$$k = r, r+1, \dots, r+m$$

Ans 6

10 pair of shoes
So 20 individual shoes.

1 Probability of no complete pair

choosing which 8 of 10 pair

$$= {}^{10}C_8$$

From each Selected pair, choose
left or right shoe

$$= 2^8$$

Favourable outcome = ${}^{10}C_8 \times 2^8$

$$P(\text{no complete pair}) = \frac{{}^{10}C_8 \times 2^8}{{}^{20}C_8}$$

$$= \frac{45(256)}{125970}$$

$$= 0.09145$$

$$\approx 9.15\%$$

ii Prob of one complete pair

$$= {}^{10}C_1 = 10$$

After Selecting both shoes
from that pair, we
need $8-2=6$
additional shoes

The other 6 shoes must
come from 6 different pair

Choose 6 pair = 9C_6

For each pair, Select one shoe
 $= 2^6$

Favourable Selection = $10 \times {}^9C_6 \times 2^6$

$P(\text{exactly 1 complete pair}) =$

$$\frac{10 \times {}^9C_6 \times 2^6}{20C_9}$$

$$\frac{10(84)(64)}{125970}$$

$$= 0.42673$$

$$\boxed{\approx 42.68\%}$$